

# RADIAL TYPE

**TK** Series

Wide Temperature Range

JAMICON®

SM ← TK → TM

- High temperature 105°C and high reliability



## SPECIFICATION

| Item                                       | Characteristic                                                                                                                                |      |                                                                                |      |      |                                        |      |      |      |        |              |         |      |         |      |     |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|------|--------------------------------------------------------------------------------|------|------|----------------------------------------|------|------|------|--------|--------------|---------|------|---------|------|-----|
| Operation Temperature Range                | -55 ~ +105°C                                                                                                                                  |      |                                                                                |      |      | -40 ~ +105°C                           |      |      |      |        | -25 ~ +105°C |         |      |         |      |     |
| Rated Working Voltage                      | 6.3 ~ 100VDC                                                                                                                                  |      |                                                                                |      |      | 160 ~ 400VDC                           |      |      |      |        | 450VDC       |         |      |         |      |     |
| Capacitance Tolerance (120Hz 20°C)         | ±20%(M)                                                                                                                                       |      |                                                                                |      |      |                                        |      |      |      |        |              |         |      |         |      |     |
| Leakage Current<br>(20°C)                  | 6.3~100 VDC      I ≤0.01CV or 4 (μA)                                                                                                          |      |                                                                                |      |      | 160~450 VDC      I ≤0.03CV +40 (μA)max |      |      |      |        |              |         |      |         |      |     |
|                                            | *Whichever is greater after 3 minutes<br>I : Leakage Current(μA)    C : Rated Capacitance(μF)    V : Working Voltage(V)                       |      |                                                                                |      |      |                                        |      |      |      |        |              |         |      |         |      |     |
| Surge Voltage<br>(20°C)                    | W.V.                                                                                                                                          | 6.3  | 10                                                                             | 16   | 25   | 35                                     | 50   | 63   | 100  | 160    | 200          | 250     | 350  | 400     | 450  |     |
|                                            | S.V.                                                                                                                                          | 8    | 13                                                                             | 20   | 32   | 44                                     | 63   | 79   | 125  | 200    | 250          | 300     | 400  | 450     | 500  |     |
| Dissipation Factor (tan δ)<br>(120Hz 20°C) | Add 0.02 per 1000 μF for more than 1000 μF                                                                                                    |      |                                                                                |      |      |                                        |      |      |      |        |              |         |      |         |      |     |
|                                            | W.V.                                                                                                                                          | 6.3  | 10                                                                             | 16   | 25   | 35                                     | 50   | 63   | 100  | 160    | 200          | 250     | 350  | 400     | 450  |     |
|                                            | tan δ                                                                                                                                         | 0.24 | 0.20                                                                           | 0.17 | 0.15 | 0.12                                   | 0.10 | 0.10 | 0.08 | 0.15   | 0.15         | 0.15    | 0.20 | 0.20    | 0.20 |     |
| Low Temperature Stability                  | Impedance ratio at 120Hz                                                                                                                      |      |                                                                                |      |      |                                        |      |      |      |        |              |         |      |         |      |     |
|                                            | Rated Voltage (V)                                                                                                                             |      | 6.3                                                                            | 10   |      | 16                                     |      | 25   |      | 35~100 |              | 160~250 |      | 350~400 |      | 450 |
|                                            | -25°C / +20°C                                                                                                                                 |      | 4                                                                              | 3    |      | 2                                      |      | 2    |      | 2      |              | 3       |      | 6       |      | 15  |
|                                            | -40°C / +20°C                                                                                                                                 |      | 10                                                                             | 8    |      | 6                                      |      | 4    |      | 3      |              | 4       |      | 10      |      | —   |
| Load Life                                  | After 2000 hours application of W.V. at +105°C, the capacitor shall meet the following limits.                                                |      |                                                                                |      |      |                                        |      |      |      |        |              |         |      |         |      |     |
|                                            | Capacitance Change                                                                                                                            |      | ≤±25% of initial value for 6.3~16 W.V., ≤±20% of initial value for 25~450 W.V. |      |      |                                        |      |      |      |        |              |         |      |         |      |     |
|                                            | Dissipation Factor                                                                                                                            |      | ≤200% of initial specified value                                               |      |      |                                        |      |      |      |        |              |         |      |         |      |     |
|                                            | Leakage current                                                                                                                               |      | ≤initial specified value                                                       |      |      |                                        |      |      |      |        |              |         |      |         |      |     |
| Shelf Life                                 | At +105°C no voltage application after 1000 hours the capacitor shall meet the limits for load life characteristics. (with voltage treatment) |      |                                                                                |      |      |                                        |      |      |      |        |              |         |      |         |      |     |

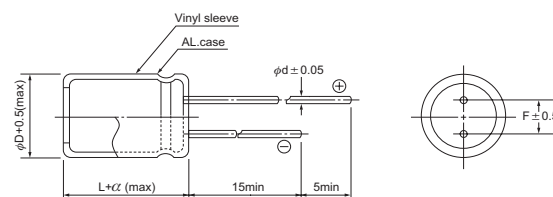
## DIMENSIONS (mm)

| $\phi D$ | 5   | 6.3 | 8   | 10  | 12.5 | 16  | 18  | 20   | 22   | 25   |
|----------|-----|-----|-----|-----|------|-----|-----|------|------|------|
| F        | 2.0 | 2.5 | 3.5 | 5.0 | 5.0  | 7.5 | 7.5 | 10.0 | 10.0 | 12.5 |
| d        | 0.5 | 0.5 | 0.6 | 0.6 | 0.6  | 0.8 | 0.8 | 0.8  | 2.0  | 1.0  |
| $\alpha$ | 1.5 | 1.5 | 1.5 | 1.5 | 1.5  | 1.5 | 1.5 | 2.0  | 2.0  | 2.0  |

## RIPPLE CURRENT COEFFICIENTS

| Temperature(°C) | 65   | 85   | 105  |
|-----------------|------|------|------|
| Multiplier      | 1.75 | 1.40 | 1.00 |

| Frequency(Hz) | 60         | 120  | 1K   | $\geq 10K$ |
|---------------|------------|------|------|------------|
| W.V.          | Multiplier |      |      |            |
| 6.3~25V       | 0.85       | 1.00 | 1.10 | 1.20       |
| 35~100V       | 0.80       | 1.00 | 1.15 | 1.25       |
| 160~250V      | 0.75       | 1.00 | 1.25 | 1.40       |
| 350~450V      | 0.70       | 1.00 | 1.30 | 1.80       |



● CASE SIZE & MAX RIPPLE CURRENT

Case size : D x L (mm)  
Max ripple current : mA(rms) 105°C 120Hz

| μF    | Code | V(Code)<br>Item | 6.3 (0J) |      | 10 (1A) |      | 16 (1C) |      |
|-------|------|-----------------|----------|------|---------|------|---------|------|
|       |      |                 | DxL      | R.C. | DxL     | R.C. | DxL     | R.C. |
| 47    | 470  |                 |          |      |         | →    | 5x11    | 75   |
| 100   | 101  |                 | 5x11     | 95   | 5x11    | 100  | 5x11    | 110  |
| 220   | 221  |                 | 5x11     | 140  | 5x11    | 150  | 6.3x11  | 190  |
| 330   | 331  |                 | 6.3x11   | 200  | 6.3x11  | 210  | 8x11.5  | 270  |
| 470   | 471  |                 | 6.3x11   | 230  | 6.3x11  | 260  | 8x11.5  | 320  |
| 1000  | 102  |                 | 8x11.5   | 400  | 10x12.5 | 460  | 10x16   | 550  |
| 2200  | 222  |                 | 10x16    | 660  | 10x20   | 790  | 12.5x20 | 910  |
| 3300  | 332  |                 | 10x20    | 860  | 12.5x20 | 990  | 12.5x25 | 1170 |
| 4700  | 472  |                 | 12.5x20  | 1040 | 12.5x25 | 1230 | 16x25   | 1310 |
| 6800  | 682  |                 | 12.5x25  | 1300 | 16x25   | 1390 | 16x31.5 | 1620 |
| 10000 | 103  |                 | 16x25    | 1450 | 16x35.5 | 1780 | 18x35.5 | 1970 |
| 15000 | 153  |                 | 16x35.5  | 1860 | 18x35.5 | 2060 | 20x40   | 2210 |
| 22000 | 223  |                 | 18x40    | 2250 | 20x40   | 2460 | 22x50   | 2940 |
| 33000 | 333  |                 | 22x50    | 2950 | 22x50   | 3020 | 25x50   | 3300 |

All blank voltage on sleeve marking is the same voltage as" → "point to.

| μF    | Code | V(Code)<br>Item | 25 (1E) |      | 35 (1V) |      | 50 (1H) |      |
|-------|------|-----------------|---------|------|---------|------|---------|------|
|       |      |                 | DxL     | R.C. | DxL     | R.C. | DxL     | R.C. |
| 0.1   | 0R1  |                 |         |      |         | →    | 5x11    | 5    |
| 0.22  | R22  |                 |         |      |         | →    | 5x11    | 7    |
| 0.33  | R33  |                 |         |      |         | →    | 5x11    | 8    |
| 0.47  | R47  |                 |         |      |         | →    | 5x11    | 10   |
| 1     | 010  |                 |         |      |         | →    | 5x11    | 15   |
| 2.2   | 2R2  |                 |         |      |         | →    | 5x11    | 22   |
| 3.3   | 3R3  |                 |         |      |         | →    | 5x11    | 27   |
| 4.7   | 4R7  |                 |         |      |         | →    | 5x11    | 32   |
| 10    | 100  |                 | 5x11    | 38   | 5x11    | 42   | 5x11    | 46   |
| 22    | 220  |                 | 5x11    | 55   | 5x11    | 65   | 5x11    | 70   |
| 33    | 330  |                 | 5x11    | 70   | 5x11    | 75   | 5x11    | 85   |
| 47    | 470  |                 | 5x11    | 80   | 5x11    | 90   | 6.3x11  | 110  |
| 100   | 101  |                 | 6.3x11  | 140  | 6.3x11  | 150  | 8x11.5  | 190  |
| 220   | 221  |                 | 8x11.5  | 240  | 8x11.5  | 260  | 10x12.5 | 300  |
| 330   | 331  |                 | 8x11.5  | 290  | 10x12.5 | 340  | 10x16   | 410  |
| 470   | 471  |                 | 10x12.5 | 360  | 10x16   | 450  | 10x20   | 540  |
| 1000  | 102  |                 | 10x20   | 650  | 12.5x20 | 770  | 12.5x25 | 930  |
| 2200  | 222  |                 | 12.5x25 | 1060 | 16x25   | 1180 | 16x35.5 | 1480 |
| 3300  | 332  |                 | 16x25   | 1240 | 16x35.5 | 1570 | 18x35.5 | 1790 |
| 4700  | 472  |                 | 16x31.5 | 1520 | 18x35.5 | 1840 |         |      |
| 6800  | 682  |                 | 18x35.5 | 1890 |         |      |         |      |
| 10000 | 103  |                 | 20x40   | 2270 |         |      |         |      |
| 15000 | 153  |                 | 22x50   | 2840 |         |      |         |      |
| 22000 | 223  |                 | 25x50   | 3210 |         |      |         |      |

● CASE SIZE & MAX RIPPLE CURRENT

Case size : D x L (mm)  
Max ripple current : mA(rms) 105°C 120Hz

| μF   | Code | V(Code)<br>Item | 63 (1J) |      | 100 (2A) |      |
|------|------|-----------------|---------|------|----------|------|
|      |      |                 | DxL     | R.C. | DxL      | R.C. |
| 0.1  | 0R1  |                 |         | →    | 5x11     | 5    |
| 0.22 | R22  |                 |         | →    | 5x11     | 8    |
| 0.33 | R33  |                 |         | →    | 5x11     | 9    |
| 0.47 | R47  |                 |         | →    | 5x11     | 11   |
| 1    | 010  |                 |         | →    | 5x11     | 16   |
| 2.2  | 2R2  |                 |         | →    | 5x11     | 24   |
| 3.3  | 3R3  |                 |         | →    | 5x11     | 30   |
| 4.7  | 4R7  |                 |         | →    | 5x11     | 35   |
| 10   | 100  |                 | 5x11    | 46   | 6.3x11   | 55   |
| 22   | 220  |                 | 5x11    | 70   | 6.3x11   | 85   |
| 33   | 330  |                 | 6.3x11  | 95   | 8x11.5   | 120  |
| 47   | 470  |                 | 6.3x11  | 110  | 10x12.5  | 160  |
| 100  | 101  |                 | 10x12.5 | 200  | 10x20    | 280  |
| 220  | 221  |                 | 10x16   | 340  | 12.5x25  | 490  |
| 330  | 331  |                 | 10x20   | 460  | 12.5x25  | 600  |
| 470  | 471  |                 | 12.5x20 | 580  | 16x25    | 720  |
| 1000 | 102  |                 | 16x25   | 940  | 18x40    | 1380 |
| 2200 | 222  |                 |         |      | 22x50    | 2260 |

All blank voltage on sleeve marking is the same voltage as" → "point to.

| μF   | Code | V(Code)<br>Item | 160 (2C) |      | 200 (2D) |      | 250 (2E) |      |
|------|------|-----------------|----------|------|----------|------|----------|------|
|      |      |                 | DxL      | R.C. | DxL      | R.C. | DxL      | R.C. |
| 0.47 | R47  |                 | 6.3x11   | 9    | 6.3x11   | 10   | 6.3x11   | 11   |
| 1    | 010  |                 | 6.3x11   | 14   | 6.3x11   | 15   | 6.3x11   | 16   |
| 2.2  | 2R2  |                 | 6.3x11   | 20   | 6.3x11   | 22   | 6.3x11   | 24   |
| 3.3  | 3R3  |                 | 6.3x11   | 25   | 6.3x11   | 26   | 8x11.5   | 34   |
| 4.7  | 4R7  |                 | 6.3x11   | 29   | 8x11.5   | 37   | 8x11.5   | 40   |
| 10   | 100  |                 | 8x11.5   | 50   | 10x12.5  | 55   | 10x16    | 70   |
| 22   | 220  |                 | 10x16    | 85   | 10x20    | 100  | 12.5x20  | 120  |
| 33   | 330  |                 | 10x20    | 120  | 12.5x20  | 130  | 12.5x20  | 150  |
| 47   | 470  |                 | 12.5x20  | 150  | 12.5x20  | 160  | 12.5x25  | 190  |
| 100  | 101  |                 | 12.5x25  | 240  | 16x25    | 260  | 16x31.5  | 310  |
| 220  | 221  |                 | 16x35.5  | 420  | 18x40    | 510  |          |      |
| 330  | 331  |                 | 18x40    | 580  |          |      |          |      |
| 470  | 471  |                 | 22x40    | 770  |          |      |          |      |
| 1000 | 102  |                 | 25x50    | 1330 |          |      |          |      |

| μF   | Code | V(Code)<br>Item | 350 (2V) |      | 400 (2G) |      | 450 (2W) |      |
|------|------|-----------------|----------|------|----------|------|----------|------|
|      |      |                 | DxL      | R.C. | DxL      | R.C. | DxL      | R.C. |
| 0.47 | R47  |                 | 8x11.5   | 11   | 8x11.5   | 11   | 10x12.5  | 11   |
| 1    | 010  |                 | 8x11.5   | 16   | 8x11.5   | 16   | 10x12.5  | 17   |
| 2.2  | 2R2  |                 | 8x11.5   | 24   | 10x12.5  | 26   | 10x20    | 30   |
| 3.3  | 3R3  |                 | 10x12.5  | 30   | 10x12.5  | 31   | 12.5x20  | 40   |
| 4.7  | 4R7  |                 | 10x12.5  | 36   | 10x16    | 42   | 12.5x20  | 47   |
| 10   | 100  |                 | 10x20    | 65   | 12.5x20  | 70   | 16x25    | 75   |
| 22   | 220  |                 | 12.5x25  | 110  | 12.5x25  | 120  | 16x31.5  | 130  |
| 33   | 330  |                 | 16x25    | 140  | 16x31.5  | 160  | 18x35.5  | 170  |
| 47   | 470  |                 | 16x35.5  | 190  | 18x35.5  | 210  |          |      |
| 100  | 101  |                 | 18x40    | 320  | 20x40    | 350  |          |      |
| 220  | 221  |                 | 22x50    | 580  |          |      |          |      |

| STANDARD NUMBER |      |      |                     |   |              |                   |                  |                                              |                   |      |                 | SPECIAL REQUIREMENTS |               |      |      |                    |
|-----------------|------|------|---------------------|---|--------------|-------------------|------------------|----------------------------------------------|-------------------|------|-----------------|----------------------|---------------|------|------|--------------------|
| 1               | 2    | 3    | 4                   | 5 | 6            | 7                 | 8                | 9                                            | 10                | 11   | 12              | 13                   | 14            | 15   |      |                    |
| S               | K    | R    | 2                   | 2 | 1            | M                 | 1                | C                                            | F                 | 1    | 1               | M                    |               |      |      |                    |
| SERIES          |      | TYPE | CAPACITANCE<br>(μF) |   | CAP.<br>TOL. | VOLTAGE<br>(W.V.) |                  | CASE SIZE<br>(φDmm) (L.mm)-Length of Al.case |                   |      |                 |                      |               |      |      |                    |
| Series          | Code | Type | Description         |   | CAP<br>(μF)  | Code              | Tolerance<br>(%) | Code                                         | Voltage<br>(W.V.) | Code | Diameter<br>(φ) | Code                 | Length<br>(L) | Code | Code | Description        |
| PS              | TH   | R    | Bulk                |   | 0.1          | OR1               | +10              | K                                            | 4                 | 0G   | 3               | A                    | 11            | 11   | W    | Without<br>Sleeve  |
| PT              | TX   |      |                     |   | 0.22         | R22               | -10              |                                              | 6.3               | 0J   | 3.8             | S                    | 11.5          | BB   |      |                    |
| CS              | WB   | P    | Taping (Ammo Pack)  |   | 0.33         | R33               | +15              | L                                            | 10                | 1A   | 4               | C                    | 12.5          | BC   | 1~9  | Customer<br>Assign |
| CR              | FS   |      |                     |   | 0.47         | R47               | -15              |                                              | 13                | 1P   | 5               | D                    | 31.5          | DB   | A~Z  |                    |
| CT              | UK   | C    | Lead Cut            |   | 1            | 010               | +20              | M                                            | 16                | 1C   | 6               | W                    | 35.5          | DF   | a~   | Brand              |
| CH              | NC   |      |                     |   | 2.2          | 2R2               | -20              |                                              | 20                | 1D   | 6.3             | E                    | 100           | 1H   |      |                    |
| CL              | LP   | F    | Lead Forming Cut    |   | 3.3          | 3R3               | +100             | P                                            | 25                | 1E   | 7               | Y                    | 110           | 1A   |      |                    |
| CF              | HP   |      |                     |   | 4.7          | 4R7               | -0               |                                              | 35                | 1V   | 8               | F                    | 115           | 1K   |      |                    |
| SV              | LS   | B    | Lead Forming Only   |   | 10           | 100               | +30              | Q                                            | 40                | 1G   | 10              | G                    | 120           | 1B   |      |                    |
| ST              | HS   |      |                     |   | 22           | 220               | -10              |                                              | 50                | 1H   | 12              | H                    | 121           | 1M   |      |                    |
| NT              | LT   | Y    | Lead Snap in        |   | 33           | 330               | +20              | R                                            | 63                | 1J   | 12.5            | I                    | 130           | 1C   |      |                    |
| SS              | HT   |      |                     |   | 47           | 470               | -0               |                                              | 80                | 1K   | 13              | J                    | 131           | 1P   |      |                    |
| SH              | HV   | W    | Snap in Terminal    |   | 100          | 101               | +50              | T                                            | 100               | 2A   | 16              | K                    | 140           | 1D   |      |                    |
| SL              | KP   |      |                     |   | 220          | 221               | -10              |                                              | 125               | 2B   | 18              | L                    | 144           | 1Q   |      |                    |
| NS              | RP   | G    | G Type Terminal     |   | 330          | 331               | +75              | U                                            | 160               | 2C   | 20              | M                    | 150           | 1E   |      |                    |
| SK              |      |      |                     |   | 470          | 471               | -10              |                                              | 180               | 2M   | 22              | N                    | 155           | 1N   |      |                    |
| SM              |      | V    | V Type Terminal     |   | 1000         | 102               | +20              | V                                            | 200               | 2D   | 25              | O                    | 157           | 1R   |      |                    |
| TK              |      |      |                     |   | 2200         | 222               | -10              |                                              | 250               | 2E   | 30              | P                    | 160           | 1F   |      |                    |
| TM              |      | S    | Screw Terminal Type |   | 3300         | 332               | +20              | H                                            | 315               | 2F   | 35              | Q                    | 170           | 1G   |      |                    |
| NK              |      |      |                     |   | 4700         | 472               | -5               |                                              | 330               | 2U   | 40              | R                    | 180           | 1I   |      |                    |
| LK              |      | M    | Surface Mount Type  |   | 10000        | 103               | +30              | F                                            | 350               | 2V   | 51              | V                    | 190           | 1J   |      |                    |
| WL              |      |      |                     |   | 22000        | 223               | -0               |                                              | 400               | 2G   | 64              | 1                    | 196           | 1S   |      |                    |
| WG              |      | E    | Horizontal Molded   |   | 33000        | 333               | +100             | W                                            | 450               | 2W   | 77              | 2                    | 215           | 1L   |      |                    |
| TL              |      |      |                     |   | 47000        | 473               | -10              |                                              | 500               | 2H   | 90              | 3                    | 236           | 1T   |      |                    |

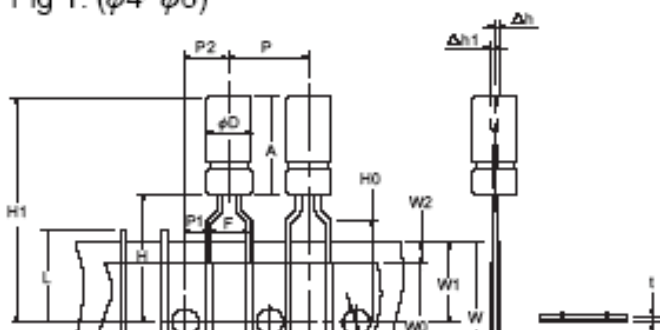
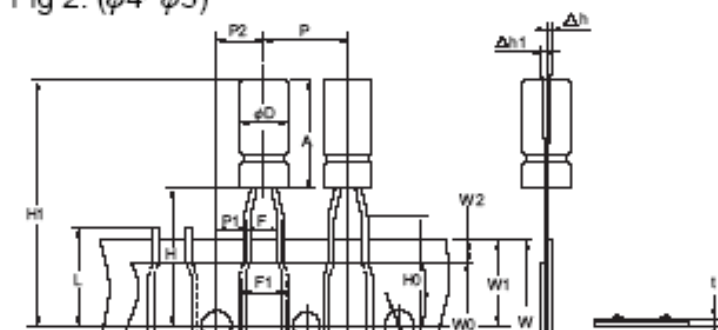
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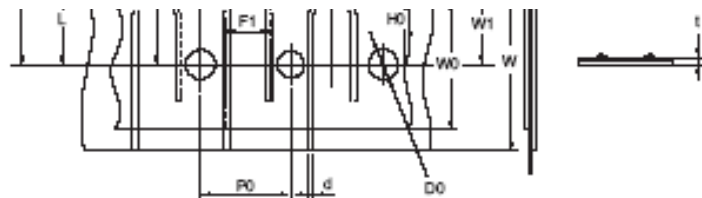
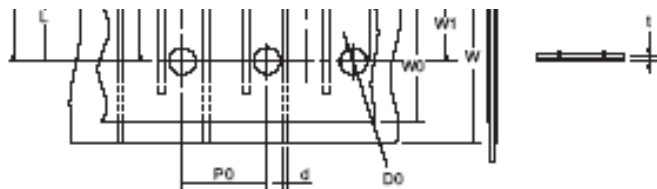
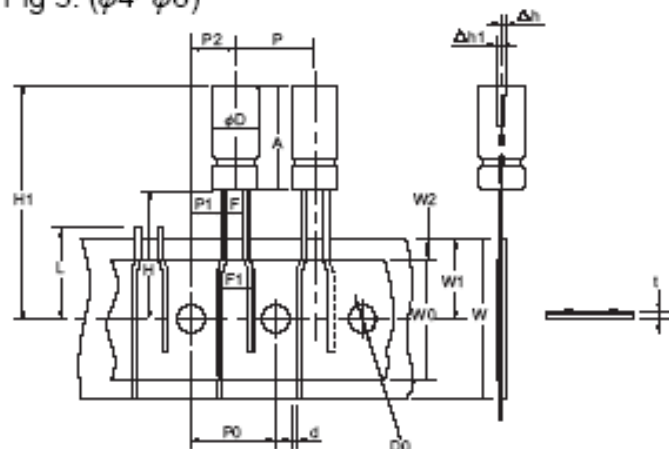
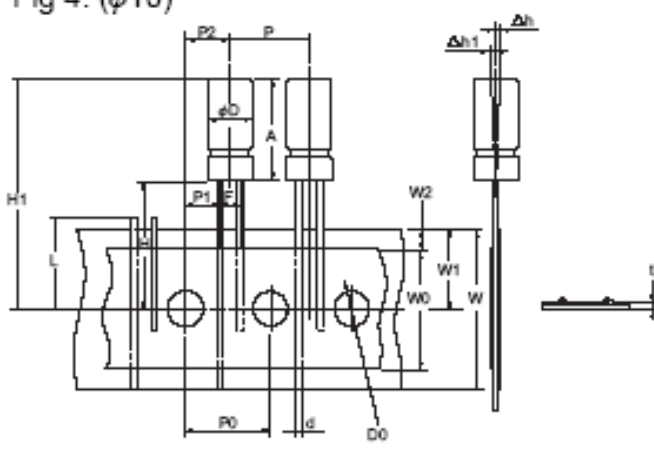
Lead taping is designed for automatic insertion equipment. Capacitor with case size of 18mm x 35.5mm or smaller are available in taping type.

## DIMENSIONS ( $\phi 4 \sim \phi 10$ )

(mm)

| Item                                         | Symbol         | Case Size |      |       |     |      |      |       |      |      |        |        |         |       |       | Tolerance    | Remark                                                                           |       |
|----------------------------------------------|----------------|-----------|------|-------|-----|------|------|-------|------|------|--------|--------|---------|-------|-------|--------------|----------------------------------------------------------------------------------|-------|
|                                              |                | 4x5       | 5x5  | 6.3x5 | 8x5 | 4x7  | 5x7  | 6.3x7 | 8x7  | 5x11 | 6.3x11 | 8x11.5 | 10x12.5 | 10x16 | 10x18 |              |                                                                                  | 10x20 |
| Lead wire diameter                           | d              | 0.45      |      |       |     |      |      |       | 0.5  |      |        | 0.6    |         |       |       | ± 0.05       |                                                                                  |       |
| Body height                                  | A              | 6.0       |      |       |     | 8.0  |      |       |      | 12.5 |        | 13     | 14      | 17.5  | 19.5  | 21.5         | max                                                                              |       |
| Intervals of bodies                          | P              | 12.7      |      |       |     |      |      |       |      |      |        |        |         |       |       | ± 1.0        |                                                                                  |       |
| Intervals of punched holes                   | P <sub>0</sub> | 12.7      |      |       |     |      |      |       |      |      |        |        |         |       |       | ± 0.2        |                                                                                  |       |
| Distance between holes and lead wire         | P <sub>1</sub> | 3.85      |      |       |     |      |      |       |      |      |        |        |         |       |       | ± 0.7        | Fig 1. Fig 4.                                                                    |       |
|                                              |                | 5.35      | 5.1  | 5.1   |     |      | 5.35 | 5.1   |      |      | 5.1    |        |         |       |       |              | Fig 2.                                                                           |       |
|                                              |                | 5.6       | 5.35 | 5.1   | 5.1 | 5.6  | 5.35 | 5.1   | 4.6  | 5.35 | 5.1    | 4.6    |         |       |       |              | Fig 3.                                                                           |       |
| Distance between holes and bodies            | P <sub>2</sub> | 6.35      |      |       |     |      |      |       |      |      |        |        |         |       |       | ± 1.0        |                                                                                  |       |
| Distance between lead and lead               | F              | 5.0       |      |       |     |      |      |       |      |      |        |        |         |       |       | +0.8<br>-0.2 | Fig 1. Fig 4.                                                                    |       |
|                                              |                | 2.0       | 2.5  | 2.5   |     |      | 2.0  | 2.5   |      |      | 2.5    |        |         |       |       |              | Fig 2. F <sub>1</sub> : 5.0 $\begin{smallmatrix} +0.5 \\ -1.0 \end{smallmatrix}$ |       |
|                                              |                | 1.5       | 2.0  | 2.5   | 2.5 | 1.5  | 2.0  | 2.5   | 3.5  | 2.0  | 2.5    | 3.5    |         |       |       |              | Fig 3. F <sub>1</sub> : 5.0 $\begin{smallmatrix} +0.5 \\ -1.0 \end{smallmatrix}$ |       |
| Base tape width                              | W              | 18.0      |      |       |     |      |      |       |      |      |        |        |         |       |       | ± 0.5        |                                                                                  |       |
| Adhesive tape width                          | W <sub>0</sub> | 12.5      |      |       |     |      |      |       |      |      |        |        |         |       |       | min          |                                                                                  |       |
| Deviation between holes and base tape        | W <sub>1</sub> | 9.0       |      |       |     |      |      |       |      |      |        |        |         |       |       | ± 0.5        |                                                                                  |       |
| Deviation between adhesive and base tape     | W <sub>2</sub> | 1.5       |      |       |     |      |      |       |      |      |        |        |         |       |       | max          |                                                                                  |       |
| Distance between body bottom and tape center | H              | 17.5      |      |       |     |      |      |       | 18.5 |      | 20.0   | 18.5   |         |       |       | ± 0.5        | Fig 1. Fig 4.                                                                    |       |
|                                              |                | 17.5      |      |       |     |      |      |       | 18.5 |      | 18.5   |        |         |       |       |              | Fig 2. Fig 3.                                                                    |       |
| Lead wire clinched hight                     | H <sub>0</sub> | 16.0      |      |       |     |      |      |       |      |      |        |        |         |       |       |              | ± 0.5                                                                            |       |
| Distance between body top and tape center    | H <sub>1</sub> | 24.5      |      |       |     | 27.5 |      |       |      | 32.5 |        |        | 33.0    | 36.0  | 38.0  | 41.0         | max                                                                              |       |
| Punched hole diameter                        | D <sub>0</sub> | 4.0       |      |       |     |      |      |       |      |      |        |        |         |       |       | ± 0.3        |                                                                                  |       |
| Length of not good lead slit                 | L              | 11.0      |      |       |     |      |      |       |      |      |        |        |         |       |       | max          |                                                                                  |       |
| Base and adhesive tape thickness             | t              | 0.6       |      |       |     |      |      |       |      |      |        |        |         |       |       | ± 0.3        |                                                                                  |       |
| Deviation of body alignment                  | Δh             | 0         |      |       |     |      |      |       |      |      |        |        |         |       |       | ± 2.0        |                                                                                  |       |
| Deviation of body alignment                  | Δh1            | 0         |      |       |     |      |      |       |      |      |        |        |         |       |       | ± 1.0        |                                                                                  |       |

Fig 1. ( $\phi 4 \sim \phi 8$ )Fig 2. ( $\phi 4 \sim \phi 5$ )

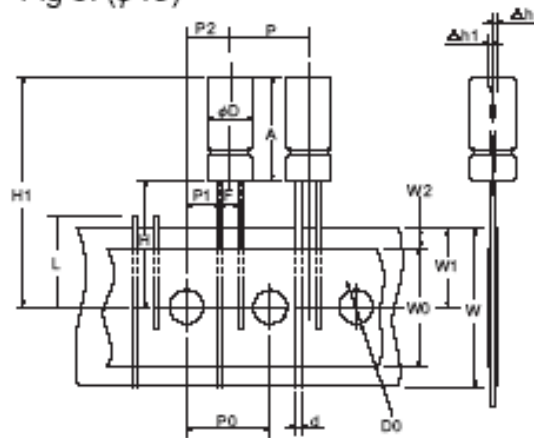
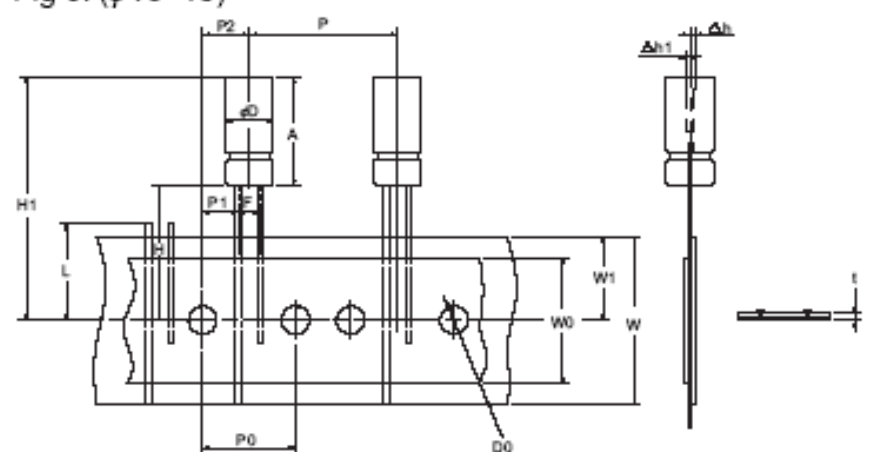
Fig 3. ( $\phi 4 \sim \phi 8$ )Fig 4. ( $\phi 10$ )

## DIMENSIONS ( $\phi 13 \sim \phi 18$ )

(mm)

| Item                                         | Symbol         | Case Size |           |           |         |           |           |           | Tolerance    | Remark        |
|----------------------------------------------|----------------|-----------|-----------|-----------|---------|-----------|-----------|-----------|--------------|---------------|
|                                              |                | 12.5 x 20 | 12.5 x 25 | 12.5 x 30 | 16 x 25 | 16 x 31.5 | 16 x 35.5 | 18 x 35.5 |              |               |
| Lead wire diameter                           | d              | 0.6       |           |           | 0.8     |           |           |           | ± 0.05       |               |
| Body height                                  | A              | 21.5      | 26.5      | 31.5      | 26.5    | 33        | 37.0      | 37.0      | max          |               |
| Intervals of bodies                          | P              | 15.0      |           |           | 30.0    |           |           |           | ± 1.0        | Fig 5. Fig 6. |
| Intervals of punched holes                   | P <sub>0</sub> | 15.0      |           |           |         |           |           |           | ± 0.2        |               |
| Distance between holes and lead wire         | P <sub>1</sub> | 5.0       |           |           | 3.75    |           |           |           | ± 0.7        |               |
| Distance between holes and bodies            | P <sub>2</sub> | 7.5       |           |           |         |           |           |           | ± 1.0        |               |
| Distance between lead and lead               | F              | 5.0       |           |           | 7.5     |           |           |           | +0.8<br>-0.2 |               |
| Base tape width                              | W              | 18.0      |           |           |         |           |           |           | ± 0.5        |               |
| Adhesive tape width                          | W <sub>0</sub> | 15.0      |           |           |         |           |           |           | min          |               |
| Deviation between holes and base tape        | W <sub>1</sub> | 9.0       |           |           |         |           |           |           | ± 0.5        |               |
| Deviation between adhesive and base tape     | W <sub>2</sub> | 1.5       |           |           |         |           |           |           | max          |               |
| Distance between body bottom and tape center | H              | 16.5      |           |           | 18.5    |           |           |           | ± 0.5        | Fig 5. Fig 6. |
| Distance between body top and tape center    | H <sub>1</sub> | 40.5      | 45.5      | 50.5      | 46.5    | 53.5      | 56.5      | 56.5      | max          |               |
| Punched hole diameter                        | D <sub>0</sub> | 4.0       |           |           |         |           |           |           | ± 0.3        |               |
| Length of not good lead slit                 | L              | 11.0      |           |           |         |           |           |           | max          |               |
| Base and adhesive tape thickness             | t              | 0.6       |           |           |         |           |           |           | ± 0.3        |               |
| Deviation of body alignment                  | Δh             | 0         |           |           |         |           |           |           | ± 2.0        |               |
| Deviation of body alignment                  | Δh1            | 0         |           |           |         |           |           |           | ± 1.0        |               |

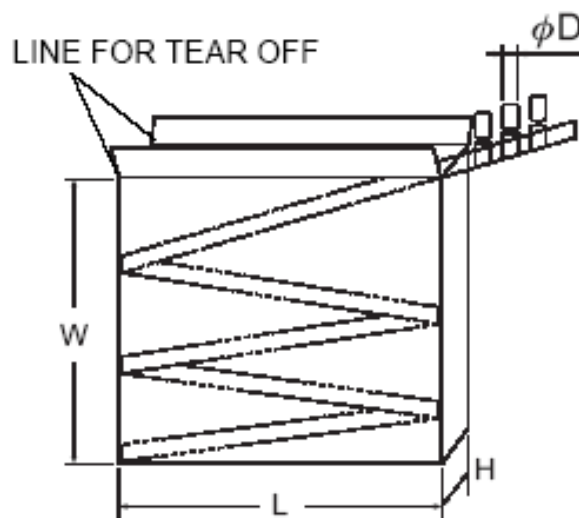
|                             |             |   |           |  |
|-----------------------------|-------------|---|-----------|--|
| Deviation of body alignment | $\Delta h$  | 0 | $\pm 2.0$ |  |
| Deviation of body alignment | $\Delta h1$ | 0 | $\pm 1.0$ |  |

Fig 5. ( $\phi 13$ )Fig 6. ( $\phi 16 \sim 18$ )

## PACKING (SYMBOL : P)

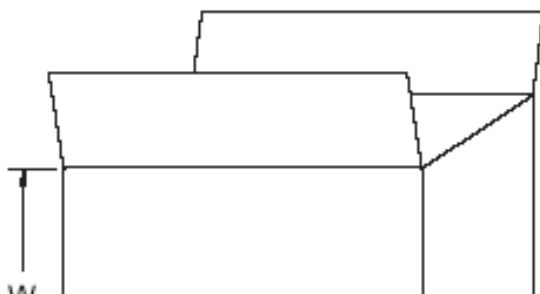
Available for various automatic equipment. Choosing the ordinal the polarity of capacitor's lead depends on customer's request.

## INNER BOX :

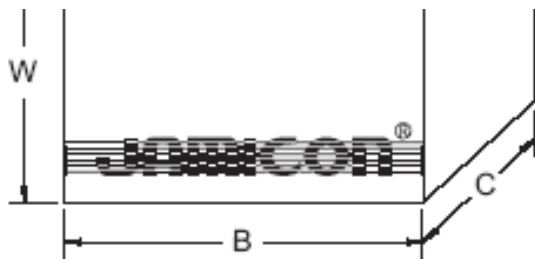


| $\phi D(\text{mm})$ | $W \pm 5(\text{mm})$ | $L \pm 5(\text{mm})$ | $H \pm 5(\text{mm})$ | Quantity(Pcs) |
|---------------------|----------------------|----------------------|----------------------|---------------|
| 4                   | 175                  | 335                  | 45                   | 2,000         |
| 5                   | 235                  | 335                  | 50                   | 2,000         |
| 6.3                 | 280                  | 335                  | 50                   | 2,000         |
| 8                   | 235                  | 335                  | 50                   | 1,000         |
| 10( $L \leq 16$ )   | 295                  | 320                  | 50                   | 800           |
| 10( $L \leq 20$ )   | 295                  | 320                  | 55                   | 800           |
| 12.5( $L \leq 20$ ) | 295                  | 320                  | 55                   | 500           |
| 12.5( $L \leq 25$ ) | 295                  | 320                  | 60                   | 500           |
| 12.5( $L \leq 30$ ) | 295                  | 320                  | 70                   | 500           |
| 16( $L \leq 25$ )   | 295                  | 320                  | 60                   | 300           |
| 16( $L \leq 31.5$ ) | 295                  | 320                  | 70                   | 300           |
| 16( $L \leq 35.5$ ) | 300                  | 320                  | 70                   | 300           |
| 18( $L \leq 35.5$ ) | 300                  | 320                  | 70                   | 243           |

## PACKING CARTON :

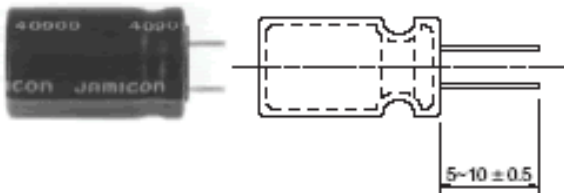
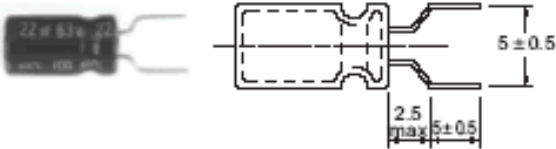
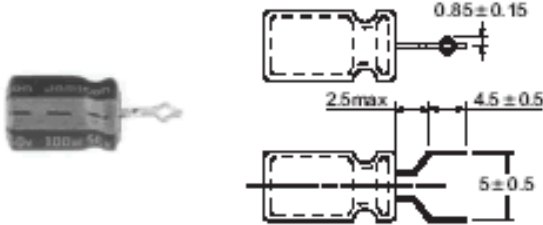
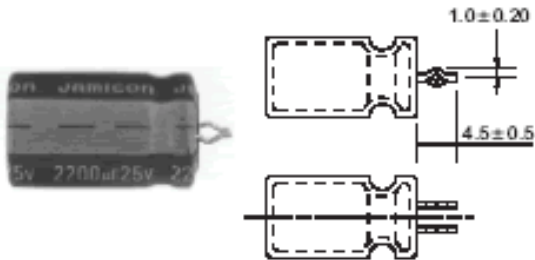


| $\phi D(\text{mm})$ | $A \pm 5(\text{mm})$ | $B \pm 5(\text{mm})$ | $C \pm 5(\text{mm})$ | Inner Box | Quantity(Pcs) |
|---------------------|----------------------|----------------------|----------------------|-----------|---------------|
| 4                   | 240                  | 355                  | 185                  | 5         | 10,000        |
| 5                   | 270                  | 355                  | 250                  | 5         | 10,000        |
| 6.3                 | 270                  | 355                  | 300                  | 5         | 10,000        |
| 8                   | 270                  | 355                  | 250                  | 5         | 5,000         |
| 10( $L \leq 16$ )   | 290                  | 345                  | 320                  | 5         | 4,000         |
| 10( $L \leq 20$ )   | 315                  | 345                  | 320                  | 5         | 4,000         |
| 12.5( $L \leq 20$ ) | 315                  | 345                  | 320                  | 4         | 2,000         |
| 12.5( $L \leq 25$ ) | 340                  | 345                  | 320                  | 4         | 2,000         |

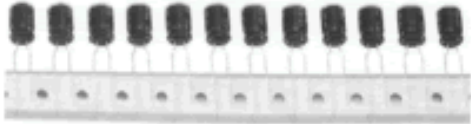
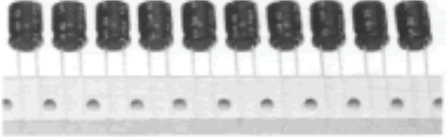


|            |     |     |     |   |       |
|------------|-----|-----|-----|---|-------|
| 12.5(L≤20) | 315 | 345 | 320 | 4 | 2,000 |
| 12.5(L≤25) | 340 | 345 | 320 | 4 | 2,000 |
| 12.5(L≤30) | 370 | 345 | 320 | 4 | 2,000 |
| 16(L≤25)   | 340 | 345 | 320 | 4 | 1,200 |
| 16(L≤31.5) | 370 | 345 | 320 | 4 | 1,200 |
| 16(L≤35.5) | 385 | 345 | 320 | 4 | 1,200 |
| 18(L≤35.5) | 385 | 345 | 320 | 4 | 972   |

## Lead Style & taping

| Item List  |                  | Code | Lead Diameter<br>(mm) | Case Size<br>DxL(mm)    | Range                  | Dimensions                                                                            |
|------------|------------------|------|-----------------------|-------------------------|------------------------|---------------------------------------------------------------------------------------|
| Lead Style | Lead Cut         | C    | 0.5~0.8               | 5 x 11<br>}<br>18 x 40  | $\phi 5 \sim \phi 18$  |    |
|            | Lead Forming Cut | F    | 0.5~0.6               | 5 x 11<br>}<br>8 x 11.5 | $\phi 5 \sim \phi 8$   |  |
|            | Snap-in          | Y    | 0.5~0.8               | 5 x 11<br>}<br>18 x 40  | $\phi 5 \sim \phi 8$   |  |
|            |                  |      |                       |                         | $\phi 10 \sim \phi 18$ |  |
|            |                  |      |                       |                         |                        | $\phi 4 \sim \phi 8$ : See Fig 1. (page 8)                                            |



|             |   |          |                               |                |                                                                                                                                                                                                                                                                                        |
|-------------|---|----------|-------------------------------|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Lead Taping | P | 0.45~0.8 | $4 \times 5$<br>}<br>18 x35.5 | $\leq \phi 18$ | <p><math>\phi 4 \sim \phi 8</math> : See Fig 1. (page 8)</p>  <p><math>\phi 10</math> : See Fig 4. (page 9)</p>  |
|-------------|---|----------|-------------------------------|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|